

Software Requirements Specification (SRS) for CuBuild

1. Introduction

1.1 Purpose

The purpose of this document is to provide a detailed specification for the CuBuild digital manufacturing marketplace, outlining its features, functionalities, and technical requirements.

1.2 Scope

CuBuild aims to address the fragmentation in the digital manufacturing industry by creating a centralized platform for designers and manufacturers to collaborate on 3D printable designs.

2. Overall Description

2.1 Product Perspective

CuBuild will serve as a standalone platform, offering a centralized hub for users involved in the digital manufacturing process. It will integrate seamlessly with external systems for data import and export.

2.2 Product Features

1. Centralized Hub for Collaboration
2. Streamlined Submission Process
3. Secure File Handling and Validation
4. Optimized Communication Channels

3. Specific Requirements

3.1 Functional Requirements

3.1.1 User Authentication

- Users must be able to register and log in securely.
- Different user roles (Designers, Manufacturers) will have specific permissions.

3.1.2 Submission Process

- Designers can submit 3D designs, including relevant information.
- Manufacturers can review and accept/reject design submissions.

3.1.3 File Handling

- Implement secure file upload and validation processes.
- Support for various 3D file formats.

3.1.4 Communication

- Real-time messaging features for designers and manufacturers to communicate.

- Notification system for updates and alerts.

3.2 Non-functional Requirements

3.2.1 Performance

- Response time for critical operations should be less than 2 seconds.
- Platform should handle concurrent user activities efficiently.

3.2.2 Security

- Use HTTPS for secure communication.
- Implement role-based access control to protect sensitive data.

3.2.3 Scalability

- The system should scale horizontally to accommodate increasing user traffic.

4. External Interface Requirements

4.1 User Interfaces

- User-friendly web interface for both designers and manufacturers.

4.2 APIs

- Define APIs for integration with external systems.

5. Constraints

- The project must adhere to data protection regulations and intellectual property laws.
- The system should be compatible with modern web browsers.

6. Conclusion

This Software Requirements Specification outlines the key features, functionalities, and technical requirements for the CuBuild digital manufacturing marketplace. It serves as a comprehensive guide for the development team, ensuring a clear understanding of project goals and user expectations.